

Washington Nationals

Software Engineering Internship

Web Development Programming Assignment

November 2025



This assignment gives you an opportunity work with a real-world baseball dataset and showcase your web development skills. It represents the kind of work done in our group and is meant to be a small but hopefully fun challenge.

We'd like you to build a web application with a front-end interface for summarizing pitch data by pitcher. The [dataset](#) for this assignment contains two files: a README and a SQLite database with pitch data from the 2025 MLB season and a player lookup.

The data is confidential. Please don't share this dataset or post your code publicly.

You are welcome to work in any open source language/framework of your choosing. We encourage you to build a single page application that uses a front-end javascript framework with a back-end API.

Requirements

1. Allow the user to select a pitcher. When a pitcher is selected, display their pitch summary table (described in #2 below). Allow the user to select one of these pitchers:

Pitcher Name
Logan Webb
Carlos Rodón
Garrett Crochet
Zac Gallen
Max Fried
Jake Irvin
MacKenzie Gore
Brad Lord
Jose A. Ferrer
Matt Waldron

2. Display a summary table of a pitcher's pitches from the 2025 season broken out by pitch type. For each pitch type thrown by the pitcher, display:
 - a. Number of pitches of this type thrown by the pitcher
 - b. Pitch usage percentage (i.e. percentage of all pitches thrown by this pitcher which are of this type)
 - c. Average pitch speed
 - d. Average horizontal break
 - e. Average induced vertical break
 - f. Average spin rate

- g. Average hit exit speed
 - h. Average hit launch angle
3. Add an "About" page to tell us a little about yourself and your submission.
 4. In zip file, include some screen shots of your app, including Brad Lord's page.
 5. Use git for version control and ensure the git directory is included in the zip file that is submitted.
 6. We'd like to run your application, so include instructions on how to run your app and what dependencies it has.
 7. Include 1-2 pages of writing (approximately 400-1000 words) explaining how you worked on the assignment. You can also include an appendix. Address the following questions:
 - a. What was your overall development process on this project?
 - i. What design decisions did you make?
 - ii. How did you determine which packages/frameworks/technology to use?
 - b. Were any of the packages/frameworks/languages used on this project new to you?
 - i. If you were already familiar with these packages/frameworks/languages, how did you learn them and use them previously?
 - c. Did you use any AI tools to complete this project?
 - i. Which AI tools did you use?
 - ii. What prompts worked well, and what prompts did not work well? Please share some prompts in an appendix.

Deliverables

Please share your code, writeup as a PDF, .git directory, screenshots, and any documentation needed to run your code.

Upload a .zip or .tar.gz file with your submission to [OneDrive](#), and provide your first and last name when uploading the file.